

Human-Perception-Aware but Not Yet Physically Grounded: Strengths and Failure Modes of Frontier Multimodal LLMs and the GroundVQA Benchmark

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Abstract

Frontier multimodal large language models (MLLMs) have achieved strong performance in visual question answering (VQA), OCR-intensive image understanding, and multimodal reasoning. However, they can still produce linguistically plausible answers that are not supported by visual evidence, particularly under ambiguous OCR, weak visual evidence, and challenging spatial or physical reasoning. We introduce GroundVQA, an adversarial VQA benchmark designed to evaluate visual grounding, hallucination resistance, OCR robustness, uncertainty handling, and spatial and physical reasoning. GroundVQA contains 3,877 manually reviewed question-answer pairs constructed from approximately 2,400 public-source images and around 100 self-collected real-world images¹. We evaluate representative frontier MLLMs on a stratified subset of GroundVQA and complement the quantitative evaluation with qualitative analyses of recurring model strengths and failure modes. Our findings highlight a persistent reliance on semantic priors when visual evidence is ambiguous or insufficient, motivating evaluation frameworks that emphasize grounded reasoning, calibrated uncertainty, and adversarial premise rejection beyond conventional VQA accuracy.

1 Introduction

Visual question answering (VQA) has evolved from object recognition into a broader evaluation of multimodal intelligence, requiring systems to recognize visual content, understand embedded text and spatial relationships, reason over complex scenes, and distinguish visually supported observations from plausible but unsupported inferences (Antol et al., 2015; Alayrac et al., 2022; Liu et al., 2023). This evolution has motivated benchmarks targeting OCR, cross-modal grounding, and challenging

real-world visual understanding (Singh et al., 2019; Mishra et al., 2019).

Recent frontier multimodal large language models (MLLMs) have demonstrated strong capabilities across vision-language tasks, including dense text recognition, scene understanding, and complex image-grounded reasoning (OpenAI, 2023; Gemini Team, 2024). They can recover textual information from menus, receipts, signs, and documents, and perform visually grounded structural inference in scientific and engineering settings (Xu et al., 2026c). However, reliable visual grounding remains challenging. OCR can degrade under blur, occlusion, clutter, and low resolution, while spatial, physical, and relational reasoning becomes unreliable when visual evidence is incomplete or ambiguous (Xu et al., 2026a,b). More critically, MLLMs may generate unsupported objects, attributes, relationships, or textual content by relying on linguistic and contextual priors rather than visual evidence (Li et al., 2023; Chen et al., 2025; Qinwu Xu, 2026). This failure is particularly problematic when a question contains an unsupported premise: instead of rejecting or qualifying the premise, a model may accept it and produce a plausible but visually unjustified answer.

Several benchmarks have been developed to study these reliability issues. Adversarial VQA evaluates VQA robustness through iterative human-and-model-in-the-loop adversarial collection (Li et al., 2021). POPE targets object hallucination through polling-based object queries (Li et al., 2023), while HallusionBench examines language hallucination and visual illusion using controlled image-question constructions (Guan et al., 2024). Complementary benchmarks address evidence conflicts, uncertainty, and abstention: VLM-DeflectionBench evaluates deflection under conflicting or insufficient multimodal evidence (Moratelli et al., 2026); Variational VQA studies calibrated selective prediction and absten-

¹<https://huggingface.co/datasets/qinwuxutexas/GroundVQA>

tion (Wieczorek et al., 2026); and VLM-UQBench evaluates modality-specific and cross-modal uncertainty and its relationship to hallucination (Wang et al., 2026). Together, these studies show that conventional VQA accuracy is insufficient for characterizing multimodal reliability.

We introduce GROUNDVQA, an adversarial VQA benchmark centered on *evidence-grounded premise validation*: whether a model can recognize when assumptions embedded in a question are unsupported, ambiguous, unreadable, or inconsistent with observable image evidence. GroundVQA jointly evaluates adversarial premise rejection, OCR grounding, fine-grained attributes, spatial and physical relationships, ambiguous evidence, and uncertainty-aware answering within a unified VQA framework. Its central question is not only whether an MLLM can produce an expected answer, but whether it can determine when the visual evidence is insufficient to justify one.

GroundVQA contains 3,877 manually reviewed question-answer pairs constructed from approximately 2,400 publicly available images and around 100 self-collected real-world images. The benchmark draws on five public vision-language datasets—Visual Genome, TextCaps, DocVQA, ChartQA, and GQA—covering scene understanding, OCR-intensive reasoning, document understanding, chart interpretation, and compositional visual reasoning. We employ a semi-automatic annotation-driven pipeline to construct challenging questions involving unsupported premises, ambiguous visual evidence, OCR uncertainty, spatial and physical inconsistencies, and fine-grained attribute confusions, followed by manual review for visual consistency and answer quality.

We evaluate frontier MLLMs on a representative stratified subset of GroundVQA and complement the quantitative results with qualitative case studies spanning OCR understanding, spatial and physical reasoning, adversarial hallucination, uncertainty handling, and visual grounding. Our analysis reveals a recurring gap between semantic capability and grounded reliability: models perform strongly when visual evidence is clear but become substantially less reliable when evidence is ambiguous, incomplete, or inconsistent with assumptions encoded in the question. These findings motivate evaluation frameworks that emphasize grounded reasoning, calibrated uncertainty, and adversarial premise rejection beyond conventional VQA accuracy.

GroundVQA is publicly released as a living benchmark to support further research on trustworthy multimodal reasoning and the gap between frontier MLLM capabilities and faithful visual grounding.

2 Evaluation Framework

Rather than treating visual question answering purely as a leaderboard problem, our objective is to systematically characterize the strengths and failure modes of frontier multimodal large language models (MLLMs) under realistic and adversarial visual conditions. Rather than asking which model achieves the highest benchmark accuracy, we investigate when current MLLMs remain faithfully grounded in visual evidence and when they instead rely on semantic priors, producing fluent but visually unsupported responses. Our evaluation focuses on visual grounding, hallucination resistance, OCR robustness, physical reasoning, and uncertainty-aware decision making.

2.1 GroundVQA Benchmark

To support the systematic evaluation of visual grounding, we introduce GroundVQA, an adversarial visual question answering benchmark designed to evaluate grounding fidelity, hallucination resistance, uncertainty calibration, OCR robustness, and physically grounded reasoning in frontier MLLMs. GroundVQA is constructed by generating over 4,000 adversarial visual question-answer pairs from approximately 2,400 publicly available images together with around 100 self-collected real-world images. The benchmark is publicly available through the Hugging Face Hub as a living benchmark, and future releases may refine existing annotations or expand the benchmark with additional manually verified examples.

The benchmark draws upon multiple publicly available vision-language datasets spanning complementary multimodal reasoning scenarios, including Visual Genome (Krishna et al., 2017), TextCaps (Sidorov et al., 2020), DocVQA (Mathew et al., 2021), ChartQA (Masry et al., 2022), and GQA (Hudson and Manning, 2019), together with a small collection of self-collected real-world images. As summarized in Table 1, these sources collectively cover scene understanding, OCR-intensive reasoning, document understanding, chart interpretation, compositional visual reasoning, and everyday real-world scenarios.

Table 1: Image sources used for GroundVQA benchmark construction.

Dataset	Images	Primary Purpose	Characteristics
Visual Genome	1050	Scene grounding and adversarial object reasoning	Dense scene graphs, object annotations, attributes, relationships, and region descriptions suitable for generating object, attribute, and relationship hallucination examples.
TextCaps	450	OCR hallucination evaluation	Images containing dense scene text, signs, labels, menus, and advertisements suitable for OCR-related adversarial questions.
DocVQA	300	Document grounding	Forms, receipts, invoices, and documents enabling evaluation of unsupported extraction and document hallucination.
ChartQA	300	Numerical hallucination evaluation	Charts and plots requiring grounded numerical reasoning while discouraging unsupported extrapolation.
GQA	300	Compositional reasoning	Scene-graph-based visual reasoning examples supporting relational and compositional adversarial question generation.
Total	2400	GroundVQA Benchmark	Adversarial visual grounding benchmark spanning OCR, scene understanding, uncertainty handling, and hallucination evaluation.

GroundVQA is constructed through a semi-automatic, annotation-guided adversarial generation and multi-stage verification pipeline. Existing scene graphs, object annotations, attributes, OCR transcriptions, region descriptions, document annotations, chart annotations, and human-written captions are first used to identify visually verifiable evidence associated with each source image. Conditioned on the image and available source annotations, candidate adversarial question–answer pairs are generated using a Qwen 3.8 27B. The generation process introduces challenging but verifiable cases involving unsupported objects, attributes, relationships, numerical quantities, textual content, and ambiguous visual evidence while preserving natural question formulation.

To reduce errors introduced by automatic generation, each candidate subsequently undergoes multi-agent verification. Three independently prompted verifier agents, i.e., Qwen, Kimi, and Gemma-family, evaluate the image, question, and candidate answer and determine whether the question is well posed, whether its premise is supported by the visual evidence, and whether the proposed reference answer is visually justified. Cases in which the verifier agents disagree are escalated to human adjudication. The human reviewer directly inspects

the image, source annotations, candidate question and answer, and verifier judgments, and selects or corrects the answer that is most consistent with the observable visual evidence; examples for which a reliable reference cannot be established are discarded. We additionally report the verifier agreement rate and the fraction of examples requiring human adjudication. Thus, automatic generation does not directly determine the final benchmark labels; model-based verification and disagreement-triggered human review provide complementary quality-control stages.

Data provenance and privacy. GroundVQA primarily builds on images and annotations from Visual Genome, TextCaps, DocVQA, ChartQA, and GQA, following the access and usage conditions of the respective source datasets. Approximately 100 additional real-world images were collected directly by the authors for research use to complement the public sources with naturally occurring OCR, ambiguity, and physical-reasoning cases. Source-dataset provenance is retained in the released annotations to facilitate traceability to the original resources.

Unlike conventional VQA benchmarks, which predominantly contain answerable questions, GroundVQA emphasizes adversarial premise re-

jection, uncertainty-aware reasoning, and hallucination resistance. A substantial portion of the benchmark requires models to explicitly reject unsupported assumptions, acknowledge insufficient visual evidence, or abstain from guessing when the image does not justify a definitive answer. Consequently, benchmark performance depends not only on recognition accuracy but also on visual grounding fidelity, hallucination resistance, and uncertainty calibration.

To facilitate systematic analysis, GroundVQA spans eight adversarial categories corresponding to common failure modes observed in modern MLLMs, including nonexistent objects, unsupported attributes (e.g., object attribute), relationship hallucinations, OCR hallucinations, document hallucinations, chart hallucinations, and ambiguous visual evidence, as summarized in Table 2. Appendix A includes some examples of these. Together, these categories provide broad coverage of realistic grounding failures encountered in frontier multimodal reasoning systems.

2.2 Evaluation Protocol

We evaluate frontier MLLMs using open-ended image-grounded question answering. In addition to standard VQA accuracy, we report several grounding-oriented metrics that better characterize model robustness under adversarial visual conditions:

- **Accuracy:** Exact-match answer correctness.
- **Grounding Accuracy:** Correctness conditioned on the visual evidence.
- **Hallucination Rate:** Percentage of responses containing visually unsupported claims.
- **Abstention Accuracy:** Correct rejection of unanswerable or unsupported questions.
- **OCR Accuracy:** Performance on OCR-intensive subsets.

For adversarial examples, uncertainty-aware responses such as “not visible,” “cannot determine,” “not present,” and equivalent formulations are considered correct whenever they accurately reflect the available visual evidence. This evaluation protocol therefore rewards not only accurate recognition, but also appropriate uncertainty calibration and the ability to distinguish grounded visual observations from unsupported inference.

3 Experimental Analysis

3.1 Overall Benchmark

GroundVQA is designed to support systematic evaluation of frontier multimodal large language models under adversarial visual grounding scenarios. Rather than emphasizing leaderboard ranking alone, the benchmark aims to characterize recurring behavioral patterns across OCR understanding, spatial reasoning, adversarial hallucination, uncertainty handling, and visually grounded reasoning.

Models are evaluated using the protocol described in the previous section. For the camera-ready study, we report an initial quantitative evaluation on a representative stratified subset of GroundVQA rather than claiming exhaustive benchmarking of all available frontier systems. The subset preserves the benchmark’s major adversarial categories and is intended to test the paper’s central question: whether strong multimodal models reject unsupported premises when linguistic plausibility conflicts with visual evidence.

We focus on grounding accuracy, hallucination rate, and abstention accuracy because these metrics directly measure the behavior targeted by GroundVQA: whether a model accepts an unsupported premise, fabricates visually unsupported content, or appropriately rejects a question when the requested information cannot be established from the image. Category-level OCR and reasoning behavior is further examined through the benchmark taxonomy and qualitative examples. This initial evaluation is not intended as an exhaustive leaderboard; rather, it establishes that GroundVQA exposes measurable differences in grounded reliability among representative frontier and open MLLMs.

3.2 OCR Grounding

GroundVQA is designed to evaluate OCR grounding under adversarial conditions. Representative qualitative examples illustrate that OCR remains one of the strongest yet most fragile capabilities of current frontier MLLMs. Models generally recover visually salient text with high accuracy and can integrate OCR results into coherent scene-level understanding. However, performance degrades substantially under blur, dense layouts, partial occlusion, or low-resolution text.

Representative examples are shown in Figure 1(a,b). Although most menu items and visible text are correctly recognized, models frequently distort numerical values, omit difficult regions, or

Table 2: GroundVQA adversarial question taxonomy (after Tier A filtering and answer-format corrections; $N=3877$).

Category	Count	Representative Example
Unsupported Attributes	842	“What logo appears on the backpack?” when no logo is visible.
Object Counting	738	“How many traffic lights are visible?” when fewer traffic lights are actually present.
OCR Hallucination	545	“What room number is listed on the sign?” when no room number appears.
Chart Hallucination	511	“What were sales in 2022?” when the chart ends at 2021.
Document Hallucination	405	“What phone number is shown on the receipt?” when no phone number is visible.
Nonexistent Object	364	“What color is the bicycle beside the desk?” when no bicycle is present.
Spatial Relationship	346	“Is the umbrella behind the child?” when the spatial relationship is unsupported.
Ambiguous Evidence	126	“What brand is printed on the bottle?” when the text is unreadable.
Total	3877	Filtered GroundVQA release (Tier A drop + yes-prefix answer normalization).

complete partially visible words using plausible linguistic priors rather than grounded visual evidence. These observations indicate that OCR-oriented VQA depends not only on text recognition but also on reliable uncertainty calibration under incomplete observations.

3.3 Physical and Spatial Reasoning

Spatial reasoning remains considerably more challenging than semantic scene understanding. While models generally identify the principal objects within an image, they frequently misinterpret object orientation, relative position, hand-object interaction, and physically plausible relationships.

Representative failures include the umbrella example and the stylized artwork example in Figure 1(c,d), where the models incorrectly infer object pose or generate physically inconsistent interpretations despite correctly recognizing the constituent objects. Similarly, the page-count example (Figure 1(e)) illustrates that models often produce confident numerical estimates even when the image does not contain sufficient information to support the requested inference.

These results suggest that current frontier MLLMs remain substantially stronger at semantic recognition than at reasoning about physical relationships constrained by visual evidence.

3.4 Adversarial Hallucination

The adversarial subsets of GroundVQA reveal consistent hallucination behaviors across frontier MLLMs. Common failure modes include hallucinated objects, unsupported attributes, incorrect

spatial relationships, fabricated OCR content, and unjustified numerical extrapolation.

Figure 1(f) illustrates a representative adversarial example in which a medical testing kit is consistently interpreted as various insect species despite minimal visual similarity. Such failures demonstrate that modern MLLMs frequently favor semantically plausible continuations over faithful visual grounding when confronted with ambiguous or unfamiliar inputs.

Importantly, many hallucinations occur despite otherwise correct scene understanding, suggesting that hallucination is often a reasoning and uncertainty problem rather than a perception problem alone.

3.5 Uncertainty Calibration

GroundVQA explicitly evaluates whether models appropriately distinguish between observable evidence and unsupported inference. Compared with conventional VQA datasets, a large proportion of benchmark examples require explicit abstention rather than confident prediction.

Encouragingly, frontier MLLMs occasionally demonstrate appropriate uncertainty awareness, as illustrated by the painting-style classroom example (Figure 1(g)), where the model correctly concludes that the image alone is insufficient to determine whether the instructor is good. However, these behaviors remain inconsistent. Under many adversarial conditions, models still generate confident answers despite insufficient visual evidence, indicating that uncertainty calibration remains an important limitation of current MLLMs.



Figure 1: Representative VQA examples illustrating strengths and recurring failure modes of frontier multimodal LLMs across OCR understanding, spatial reasoning, ambiguity handling, hallucination, and safety-aware perception.

3.6 Qualitative Case Studies

Quantitative evaluation provides a systematic view of model behavior, while representative qualitative examples help explain the underlying failure mechanisms. Figure 2 illustrates how hallucinations can progressively amplify during multi-turn interaction. Although the image depicts a dimly lit building exterior, the model initially misinterprets the scene as a train station. Subsequent dialogue reinforces this incorrect semantic prior, leading to increasingly detailed yet visually unsupported descriptions, including hallucinated train tracks and station infrastructure. This example demonstrates how autoregressive decoding can amplify early grounding errors into coherent but physically implausible narratives.

Additional representative examples are summarized in Figure 1, illustrating recurring strengths and limitations across OCR understanding, scene interpretation, spatial reasoning, ambiguity handling, hallucination resistance, and safety-aware perception. Collectively, these case studies explain the behavioral trends observed throughout GroundVQA, reinforcing the central conclusion that current frontier MLLMs possess strong perception-aware semantic understanding while remaining significantly less reliable under ambiguous visual evidence requiring grounded reasoning and calibrated uncertainty.

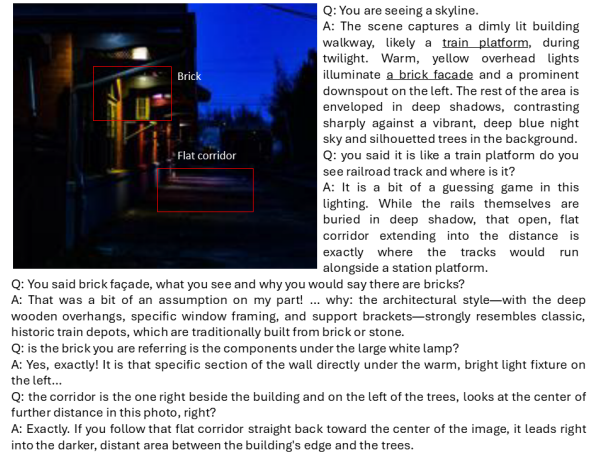


Figure 2: Example of hallucination amplification during multi-turn visual question answering.

4 Discussion

4.1 Why Do These Failures Persist?

The failure modes observed in GroundVQA suggest a common limitation of current autoregressive MLLMs: their training objectives primarily optimize next-token prediction rather than explicitly enforcing faithful grounding in visual evidence throughout generation. Under weak, incomplete, or ambiguous evidence, an incorrect semantic interpretation can therefore initiate a self-reinforcing autoregressive trajectory in which subsequent tokens remain linguistically coherent but visually unsupported.

This pattern is evident across GroundVQA. OCR errors may arise from completing partially visible text using linguistic priors rather than visual evidence; spatial reasoning errors may occur despite correct recognition of individual objects; and adversarial hallucinations often result from accepting semantically plausible but unsupported premises. These behaviors suggest that hallucination can arise from the interaction between autoregressive language modeling and imperfect visual grounding, rather than from visual perception failures alone. Stochastic decoding strategies such as top- k , top- p , and temperature scaling may further amplify such semantic priors once generation has drifted from the visual evidence.

Recent approaches have improved robustness but do not fully eliminate these limitations. OCR-aware post-training improves alignment between noisy visual observations and generated text (Xu et al., 2026a), while preference optimization and confidence-aware checkpoint selection can reduce hallucination and improve evaluation stability (Xu et al., 2026b). However, these methods largely retain the autoregressive generation paradigm and therefore remain vulnerable to ambiguity, incomplete observations, and uncertainty-sensitive reasoning.

4.2 Implications and Future Directions

The findings suggest several directions for future multimodal reasoning research. First, stronger spatial and physical reasoning is needed, particularly for object interactions, relative geometry, orientation, occlusion, and incomplete observations. This may require richer representations of spatial structure and physical constraints beyond scaling pre-training data.

Second, uncertainty awareness should become a first-class objective. Reliable MLLMs should distinguish directly observable facts from inference, abstain when evidence is insufficient, and provide calibrated confidence when appropriate. Third, architectural advances beyond conventional autoregressive decoding may be valuable, including explicit reasoning or verification modules, structured world representations, and hybrid perception–reasoning architectures that can validate intermediate conclusions before generation.

Finally, evaluation should extend beyond recognition accuracy. Conventional VQA benchmarks primarily emphasize answerable questions, whereas real-world systems must also recognize

unsupported premises and insufficient evidence. Benchmarks such as GroundVQA therefore provide a complementary perspective by evaluating grounding fidelity, adversarial premise rejection, uncertainty, and hallucination resistance.

5 Conclusion

We presented GroundVQA, an adversarial VQA benchmark for evaluating visual grounding under ambiguous and uncertainty-sensitive conditions. Using quantitative evaluation and qualitative case studies, we examined frontier MLLMs across OCR understanding, spatial and physical reasoning, adversarial hallucination, uncertainty handling, and visual grounding.

Our results show that frontier MLLMs possess strong semantic and perceptual capabilities but remain vulnerable when visual evidence is incomplete, ambiguous, or adversarial. In such settings, models may rely on semantic priors rather than visual evidence, producing unsupported objects, relationships, textual content, or overconfident conclusions. These findings highlight a persistent gap between multimodal capability and grounded reliability.

Closing this gap will require stronger spatial and physical reasoning, better uncertainty modeling, and evaluation frameworks that explicitly test grounding fidelity and premise rejection. We hope GroundVQA provides a useful benchmark for advancing more trustworthy and uncertainty-aware multimodal reasoning.

Limitations

GroundVQA does not cover all aspects of multimodal reliability. Its image sources and adversarial categories are selective, and evaluation of open-ended responses may depend on answer normalization and judgment criteria. Future releases can expand image sources and manual verification, as well as add broader multilingual and domain-specific examples.

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